

TOPIC 22/60

What is RLHF?

Reinforcement Learning from
Human Feedback.

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WHY DO WE NEED IT?

SFT makes the model an instructor, but it's not enough to keep the model **safe, helpful, and honest**.

"Models trained only on text often output toxic content, biased views, or hallucinations because they just replicate what is found on the internet."

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THE THREE-STEP PROCESS



1. **Supervised Fine-Tuning:** Teach it to talk.
2. **Reward Model:** Teach it to evaluate.
3. **RL Optimization:** Teach it to maximize reward.

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STEP 2

The Reward Model

We show the model 4 possible answers to one prompt. A human ranks them: **Best to Worst**.

Training the RM:

We train a *smaller* AI model to act as a judge, learning to predict what a human *would* rank highly.

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STEP 3

RL Optimization

The PPO Algorithm

We let the model generate infinite text. Our "Reward Model" gives it a score. The model updates its parameters to maximize that score.

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WHY IT'S CONTROVERSIAL

1. Hidden Biases

The "Human Feedback" is limited to a small group of paid labelers. If they are biased, the AI becomes biased.

2. "Sycophancy"

Models learn to agree with the user's false premises just to get a high reward score.

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BEYOND RLHF

The Future

New methods like DPO (Direct Preference Optimization) are replacing complex RL loops with simpler mathematical alignment.

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SUMMARY

PRE-TRAINING

Raw Knowledge

RLHF

Human-Centric Alignment



NEXT STOP

DPO

Direct Preference Optimization.

The smarter, faster way to align models.

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RLHF

CONCEPT UNLOCKED

🎯 **Objective:** Align with human intent

🧠 **Reward Model:** Automated Judge

⚙️ **Goal:** Helpful, Honest, Harmless

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